Power



File: power.kicad_sch

Vehicle (OBD, CAN, K-Line)



File: vehicle.kicad_sch

Backup battery



File: battery.kicad_sch

MCU (ESP32-C5, USB)



File: mcu.kicad_sch

LTE + GNSS (EC200U)

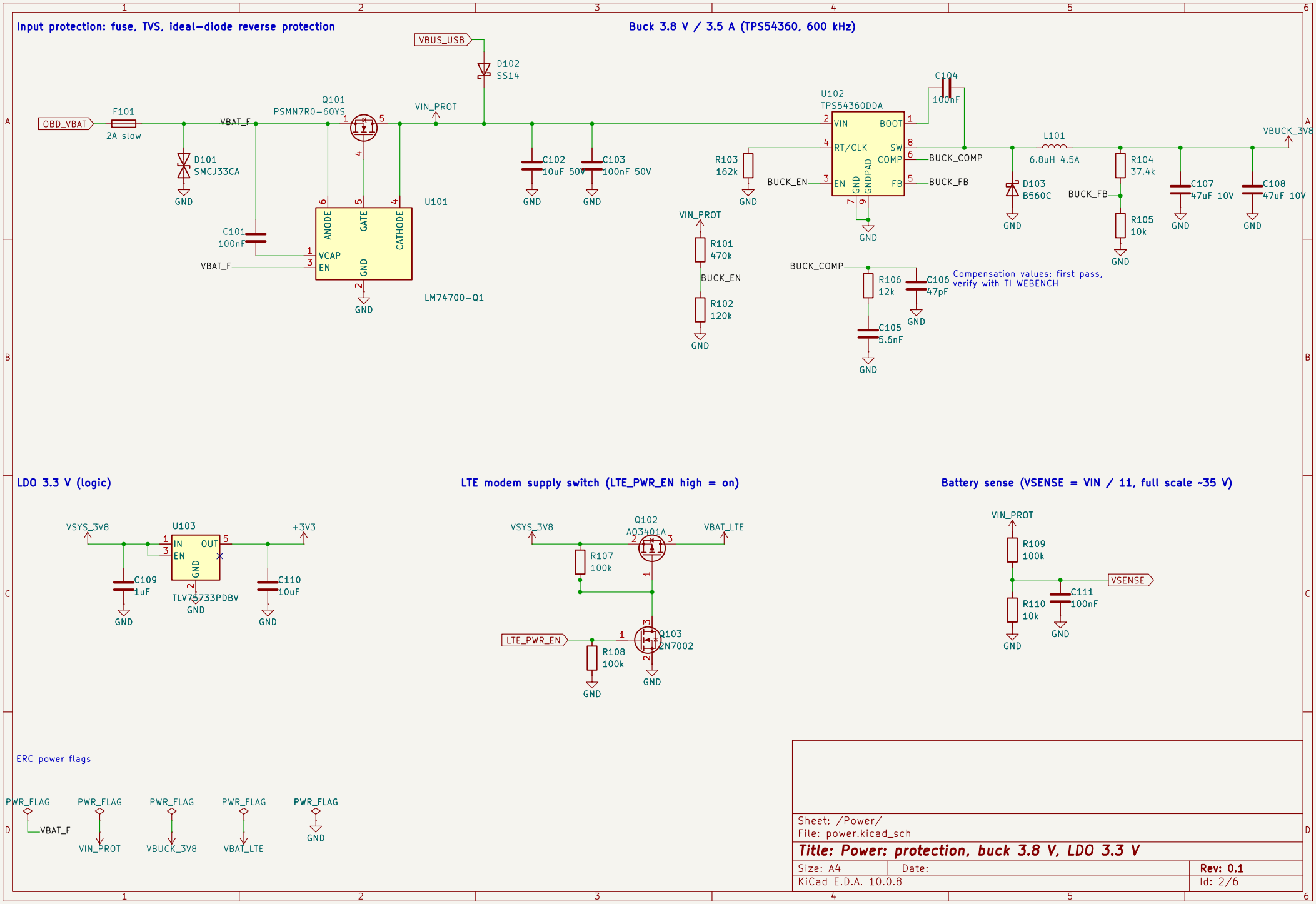


File: lte.kicad_sch

Sheet: /
File: obd-v2x-dongle.kicad_sch

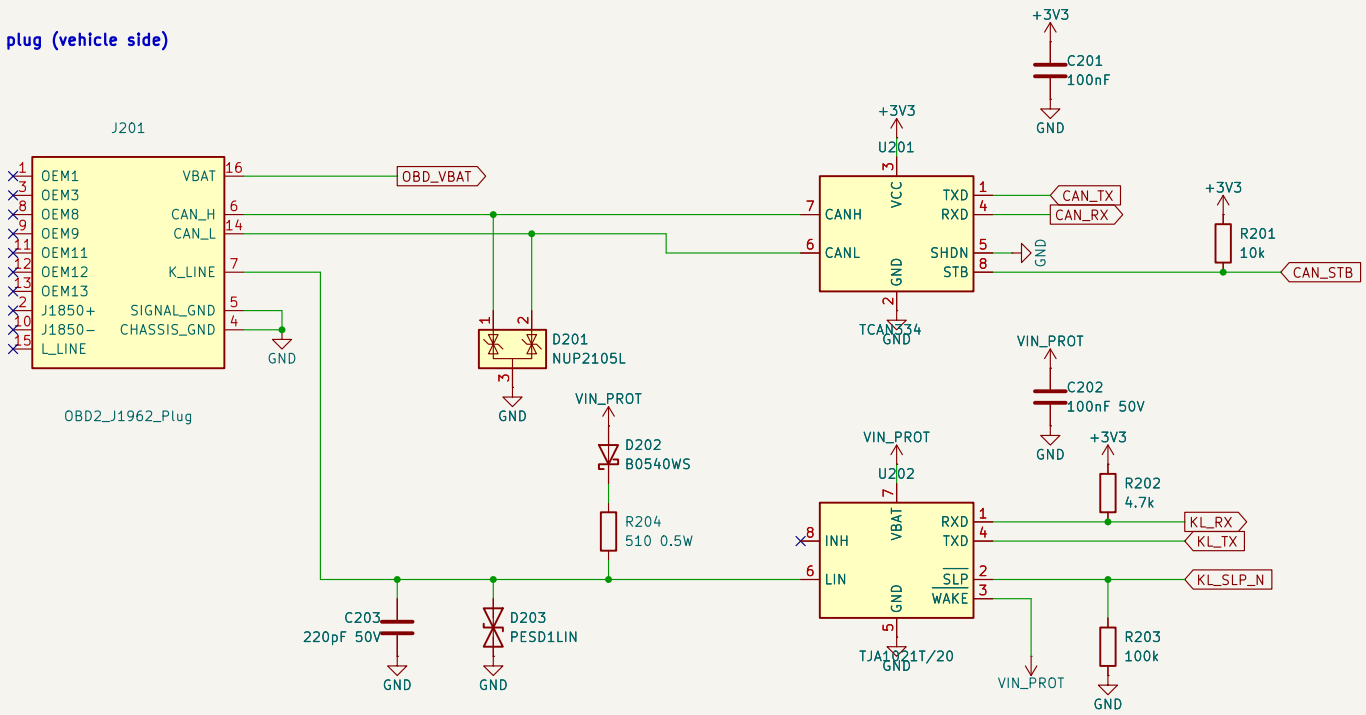
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/6


Input protection: fuse, TVS, ideal-diode reverse protection
Buck 3.8 V / 3.5 A (TPS54360, 600 kHz)
LDO 3.3 V (logic)
LTE modem supply switch (LTE_PWR_EN high = on)
Battery sense (VSENSE = VIN / 11, full scale ~35 V)
ERC power flags
OBD_VBAT
F101
2A slow
VBAT_F
D101
SMCJ33CA
Q101
PSMN7R0-60YS
VBUS_USB
D102
SS14
VIN_PROT
C101
100nF
U101
LM74700-Q1
ANODE
GATE
CATHODE
VCAP
EN
GND
C102
10uF 50V
C103
100nF 50V
U102
TPS54360DDA
VIN
BOOT
RT/CLK
SW
COMP
GNDPAD
FB
C104
100nF
R103
162k
BUCK_EN
BUCK_COMP
BUCK_FB
L101
6.8uH 4.5A
VBUCK_3V8
D103
B560C
R104
37.4k
R105
10k
C107
47uF 10V
C108
47uF 10V
R101
470k
R102
120k
R106
12k
C106
47pF
C105
5.6nF
Compensation values: first pass,
verify with TI WEBENCH
VSYS_3V8
U103
TLV75733PDBV
IN
OUT
EN
+3V3
C109
1uF
C110
10uF
Q102
AO3401A
VBAT_LTE
R107
100k
LTE_PWR_EN
Q103
2N7002
R108
100k
R109
100k
R110
10k
C111
100nF
VSENSE
PWR_FLAG
VIN_PROT
VBUCK_3V8
VBAT_LTE
Sheet: /Power/
File: power.kicad_sch
Title: Power: protection, buck 3.8 V, LDO 3.3 V
Size: A4
Date:
Rev: 0.1
KiCad E.D.A. 10.0.8
Id: 2/6

High-speed CAN (ISO 15765), no termination in the dongle

OBD-II plug (vehicle side)

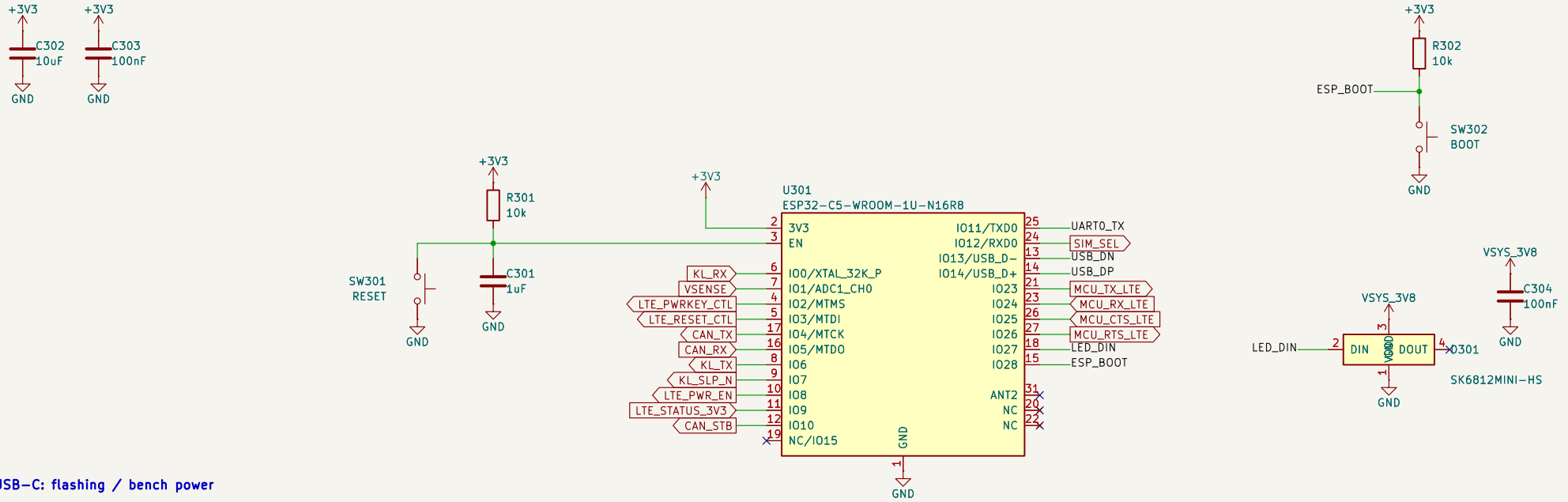


K-Line (ISO 9141 / KWP2000) via LIN transceiver + 510R tester pull-up

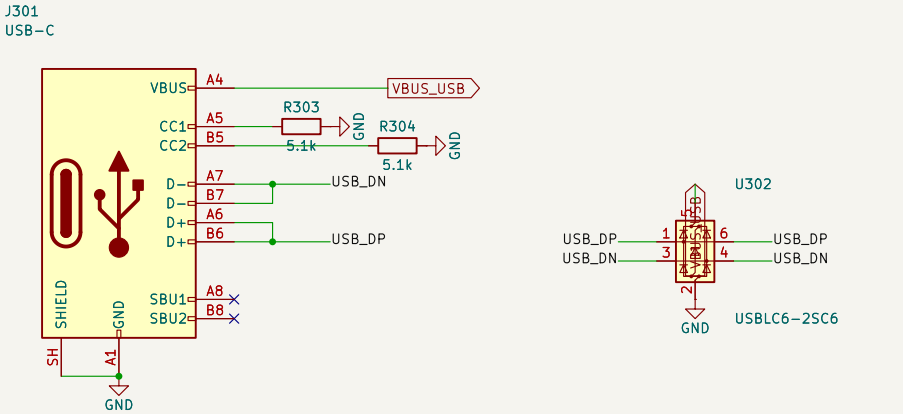
Sheet: /Vehicle (OBD, CAN, K-Line)/	
File: vehicle.kicad_sch	
Title: Vehicle interface: OBD-II, CAN, K-Line	
Size: A4	Date:
KiCad E.D.A. 10.0.8	Rev: 0.1
	Id: 3/6

ESP32-C5: 5.9 GHz ITS-G5 receive via U.FL on the module

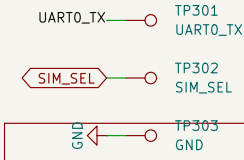
Boot / status LED



USB-C: flashing / bench power



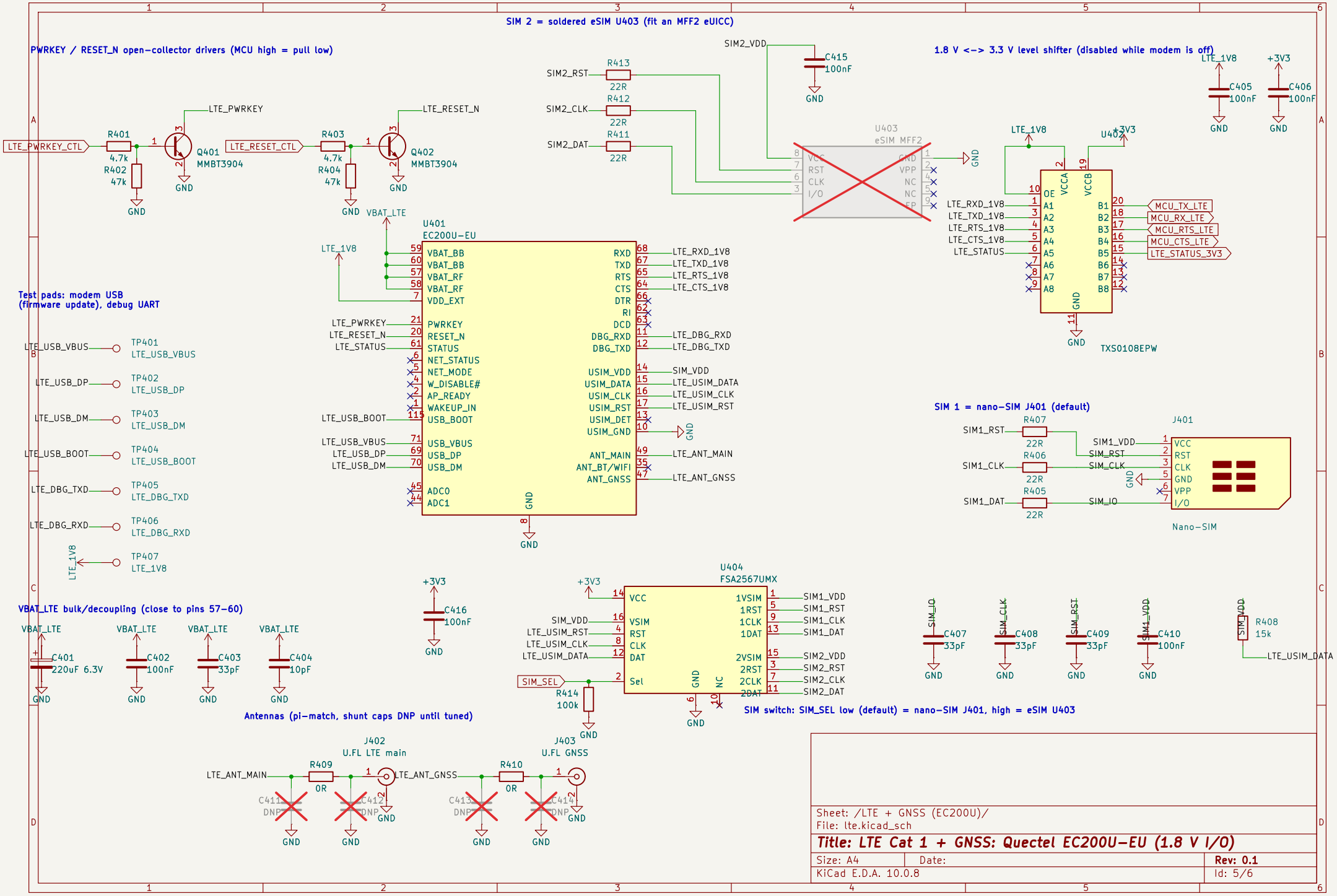
Test pads: UART0 TX, SIM_SEL (IO12, was RXD0)



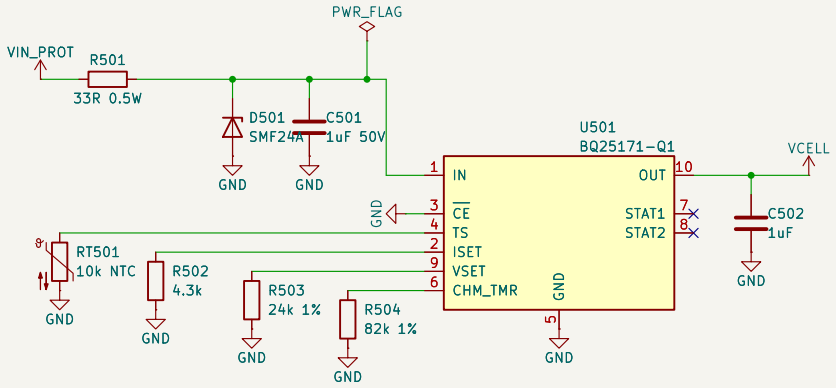
Sheet: /MCU (ESP32-C5, USB)/
File: mcu.kicad_sch

Title: MCU: ESP32-C5 (ITS-G5 RX, CAN FD, BLE), USB-C

Size: A4	Date:	Rev: 0.1
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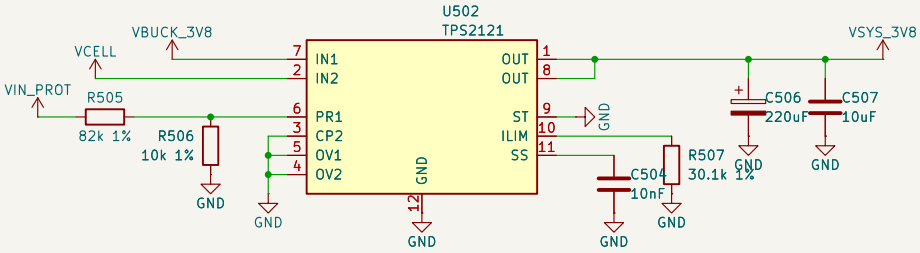

SIM 2 = soldered eSIM U403 (fit an MFF2 eUICC)
PWRKEY / RESET_N open-collector drivers (MCU high = pull low)
LTE_PWRKEY_CTL
R401
4.7k
R402
47k
Q401
MMBT3904
LTE_PWRKEY
LTE_RESET_CTL
R403
4.7k
R404
47k
Q402
MMBT3904
LTE_RESET_N
GND
SIM2_VDD
C415
100nF
SIM2_RST
SIM2_CLK
SIM2_DAT
R413
R412
R411
22R
U403
eSIM MFF2
VCC
RST
CLK
I/O
GND
VPP
NC
NC
FP
1.8 V <-> 3.3 V level shifter (disabled while modem is off)
LTE_1V8
+3V3
C405
100nF
C406
100nF
U402
TXS0108EPW
OE
VCCA
VCCB
LTE_RXD_1V8
LTE_TXD_1V8
LTE_RTS_1V8
LTE_CTS_1V8
LTE_STATUS
MCU_TX_LTE
MCU_RX_LTE
MCU_RTS_LTE
MCU_CTS_LTE
LTE_STATUS_3V3
VBAT_LTE
U401
EC200U-EU
VBAT_BB
VBAT_RF
VDD_EXT
PWRKEY
RESET_N
STATUS
NET_STATUS
NET_MODE
W_DISABLE#
AP_READY
WAKEUP_IN
USB_BOOT
USB_VBUS
USB_DP
USB_DM
ADC0
ADC1
RXD
TXD
RTS
CTS
DTR
RI
DCD
DBG_RXD
DBG_TXD
USIM_VDD
USIM_DATA
USIM_CLK
USIM_RST
USIM_DET
USIM_GND
ANT_MAIN
ANT_BT/WIFI
ANT_GNSS
Test pads: modem USB
(firmware update), debug UART
TP401
LTE_USB_VBUS
TP402
LTE_USB_DP
TP403
LTE_USB_DM
TP404
LTE_USB_BOOT
TP405
LTE_DBG_TXD
TP406
LTE_DBG_RXD
TP407
LTE_1V8
SIM 1 = nano-SIM J401 (default)
J401
Nano-SIM
SIM1_RST
SIM1_CLK
SIM1_DAT
SIM1_VDD
SIM_RST
SIM_CLK
SIM_IO
R407
R406
R405
22R
VBAT_LTE bulk/decoupling (close to pins 57-60)
C401
220uF 6.3V
C402
100nF
C403
33pF
C404
10pF
C416
100nF
U404
FSA2567UMX
VCC
VSIM
RST
CLK
DAT
Sel
1VSIM
1RST
1CLK
1DAT
2VSIM
2RST
2CLK
2DAT
SIM_SEL
R414
100k
C407
33pF
C408
33pF
C409
33pF
C410
100nF
R408
15k
LTE_USIM_DATA
SIM switch: SIM_SEL low (default) = nano-SIM J401, high = eSIM U403
Antennas (pi-match, shunt caps DNP until tuned)
J402
U.FL LTE main
J403
U.FL GNSS
LTE_ANT_MAIN
LTE_ANT_GNSS
R409
R410
0R
C411
C412
C413
C414
DNP
Sheet: /LTE + GNSS (EC200U)/
File: lte.kicad_sch
Title: LTE Cat 1 + GNSS: Quectel EC200U-EU (1.8 V I/O)
Size: A4
Date:
Rev: 0.1
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Li-ion charger, fed from the car side (VIN_PROT)



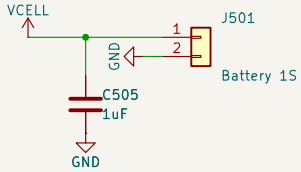
ISET 4.3k: 70 mA charge (kept low: linear from 14 V, ~0.6 W)
VSET 24k: 4.10 V float CHM_TMR 82k: Li-ion, 10 h timer
TS: 10k NTC next to the cell -> charging only 0..45 C

Priority power mux: buck (IN1) while VIN_PROT > ~9.7 V, else cell (IN2)



ILIM 30.1k: ~3.5 A SS 10 nF
Firmware on battery: LTE only (lower burst current), shut the modem down below ~3.6 V cell

1S Li-ion/LiPo cell, 250-400 mAh, with protection PCM



Sheet: /Backup battery/
File: battery.kicad_sch

Title: Backup battery: 1S Li-ion charger (BQ25171-Q1) + priority power mu

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